

Semi-Supervised Multi-Task Deep Learning for WiFi Fingerprint Database Construction in Building-Scale Localization

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Abstract—WiFi-based indoor positioning has emerged as a crucial technology for enabling smart consumer electronic applications, particularly in large-scale buildings. The construction of WiFi fingerprint databases using received signal strength (RSS) is foundational due to its widespread deployment. However, achieving high positioning accuracy typically requires labor-intensive and time-consuming site surveys. While recent crowdsourcing methods have facilitated the collection of numerous RSS samples, these samples frequently lack labels and reliability in multi-scale building environments. In this paper, we design a novel semi-supervised and multi-task mean-teacher model (MTMT-DNN) to annotate crowdsourcing unlabeled multi-scale fingerprint samples. This method enables the construction of a comprehensive fingerprint database without requiring intensive manual effort or compromising positioning accuracy. Our key idea is to first develop a multi-task Deep Neural Network (MT-DNN) for simultaneously annotating building, floor, and intra-floor coordinate labels by leveraging their complementary information. Then we employ the mean-teacher semi-supervised learning to leverage additional unlabeled fingerprint data for further improving the annotating performance and reducing intensive manual effort. Finally, we train the MTMT-DNN model by developing two multi-task loss functions and ensuring consistency between them, thereby enhancing the reliability of the annotated crowdsourced fingerprints. We conducted real-world experiments in a 20,000 m^2 site encompassing three multi-story buildings. The results demonstrate that our proposed method significantly reduces the workload of manually collecting labeled fingerprint samples. With only 20% of labeled fingerprints collected, we achieve 99% average annotation accuracy for building and floor labels and an average coordinates annotation error within 4 m .

Index Terms—Indoor localization, multi-building and multi-floor determination, WiFi RSS, multi-task deep learning, semi-supervised learning.

I. INTRODUCTION

LOCATION information is essential for many commercial positioning services in consumer electronics [1], [2]. It is required to determine the location of target devices in various consumer scenarios, such as shopping malls, airports, factories, campuses, and other large-scale urban public buildings. For instance, precise information about the building, floor, and position significantly reduces the search area for customers seeking a specific retail shop within a large shopping mall. Fortunately, the rapid development of electronic products, such as smartphones and smart wearable devices, has opened new possibilities for indoor positioning research in multi-building and multi-floor consumer electronics, as illustrated in Fig. 1.

Significant advancements in electronic devices have improved target positioning in multi-building and multi-floor environments, enhancing accuracy and reliability. Traditional global positioning system (GPS) perform well in outdoor environments, but their effectiveness is significantly reduced indoors due to obstructions from building structures and materials. However, wireless indoor positioning systems utilizing technologies such as wireless sensor networks, WiFi, Bluetooth, radio frequency identification device (RFID), and ultra-wideband (UWB) have rapidly advanced. This progress is driven by diverse consumer electronic applications, including location-based services, navigation, emergency response, and asset tracking [3], [4]. These systems can leverage sensors and communication modules in smartphones and smart wearable devices to achieve precise user positioning. Among various technologies, WiFi-based indoor positioning, particularly using received signal strength (RSS), has gained significant adoption due to the ubiquitous availability of WiFi infrastructure and the widespread popularity of mobile electronic devices [5], [6].

Recently, crowdsourcing has emerged as a promising solution to mitigate the cumbersome process of manual site surveys for WiFi RSS fingerprint indoor localization [7], [8], [9], [10]. By leveraging the widespread availability of smart devices, crowdsourcing facilitates the continuous and spontaneous collection of RSS samples from users moving within

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Fig. 1. Indoor multi-building and multi-floor positioning for consumer electronics scenarios.

Active crowdsourced labeled fingerprints	Original fingerprints database					
	AP index (B,F,A,N)	1	2	3	4	511
	(1,1,1,1)	-15db	-39db	-65db	-71db	...
	?	-20db	-35db	-70db	-68db	...
	(1,2,6,1)	-65db	-72db	-18db	-30db	...
	?	-55db	-30db	-35db	-18db	...
	?	-60db	-70db	-49db	-58db	...
	(2,3,4,3)	-78db	-82db	-32db	-43db	...
	?	-100db	-100db	-55db	-30db	...
	?	-95db	-90db	-100db	-92db	...
	...	...	...	...	...	...
	?	-46db	-51db	-65db	-79db	...

Fig. 2. There is an imbalance between labeled and unlabeled fingerprints in a multi-building and multi-floor crowdsourced database.

WiFi-covered areas. This approach significantly increases the volume of collected RSS data and keeps the fingerprint map up-to-date, thereby enhancing the positioning system's adaptability to dynamic environments.

However, crowdsourced RSS fingerprint data present challenges in multi-building and multi-floor consumer electronics scenarios. They often comprise a limited number of labeled fingerprints and a substantial quantity of unlabeled fingerprints, particularly in multi-scale environments (as illustrated in Fig. 2). Labeling these fingerprints is essential for constructing a fingerprint database and can be achieved through interpolation schemes, semi-supervised annotation, or trajectory matching methods [11], [12], [13], [14].

Current research in this area faces several significant obstacles. One major issue is the limited positioning accuracy due to the sparse density of reference points obtained through fingerprint collection. This problem arises from the time-consuming and labor-intensive nature of data gathering. Another significant challenge is the reliability of unlabeled fingerprints. The vast number of unlabeled fingerprints compared to the scarce labeled ones makes accurate annotation difficult. Additionally, existing fingerprint labeling methods primarily focus on horizontal scales and overlook building and floor scale information, leading to significant errors in positioning label annotations and accumulating errors across buildings and floors. Therefore, it is crucial to explore effective annotation techniques that can utilize a small set of labeled fingerprints to accurately label a large collection of unlabeled crowdsourced fingerprints.

In this paper, we address these issues by developing a novel approach that leverages semi-supervised learning and

multi-task learning techniques. Our goal is to facilitate the automatic annotation of unlabeled multi-scale fingerprint samples obtained through crowdsourcing, thereby contributing to the efficient construction of a fingerprint database without compromising positioning accuracy. The main contributions are summarized as follows:

- We develop a multi-task deep neural network (MT-DNN) capable of addressing position labeling tasks across multiple scales. This model can be extended to various environments with multiple buildings and floors, as it simultaneously learns two multi-class classification tasks and one regression task for accurate building, floor, and coordinates labeling. It explores the correlation between these tasks and constrains the accumulation of annotation errors across different scales.
- We propose the multi-task mean-teacher learning model (MTMT-DNN) to enhance the model's ability to generalize to new data. The MT-DNN labeling model serves as both the student and teacher network within this model, and can efficiently utilize both labeled and unlabeled fingerprints for training. Leveraging self-ensembling techniques, our MTMT-DNN model reduces the manual effort involved in constructing the fingerprint database by semi-supervisedly predicting multi-scale labels.
- We design a multi-task supervised loss function for labeled data and a multi-task consistency loss function for unlabeled data to improve the reliability of the predicted labels. This approach effectively utilizes unlabeled samples in the semi-supervised network's parameter learning process, ensuring accurate annotation of unlabeled fingerprints.

The remainder of this paper is organized as follows. Section II reviews prior research. Section III presents a problem formulation of multi-building and multi-floor fingerprint labeling. In Section IV, we present a detailed description of the key components of MTMT-DNN, including a multi-task deep learning model for fingerprint labeling, a multi-task supervised loss for labeled fingerprints, and a multi-task consistency loss for unlabeled fingerprints. Section V discusses the experimental setups and results of MTMT-DNN, along with an evaluation and comparison with existing works. Section VI gives a comprehensive discussion of our proposed MTMT-DNN and outlines directions for future work. Finally, Section VII concludes the paper with a summary.

II. RELATED WORK

In this section, we introduce existing techniques for constructing a fingerprint database with minimal manual effort from the following three perspectives: interpolation, semi-supervised learning, and trajectory matching.

A. Interpolation Schemes

Interpolation schemes utilize labeled data, including location coordinates and corresponding RSS data, to construct a position fingerprint database. These schemes rely on signal propagation models and a limited set of labeled data to perform interpolation. Although a novel radio map interpolation

scheme has been proposed to generate virtual samples and fill the sparse fingerprint database [15], but it does not account for the characteristics of signal propagation. Gaussian Process (GP) models [16] and linear interpolation [17] have also been applied to interpolate RSS measurements between reference points (RPs). Additional methods, such as those based on LASSO regression [18] or a propagation model [19] have been introduced for fingerprint interpolation. However, due to the highly dynamic and challenging nature of wireless channels in indoor environments, their performance is suboptimal in large-scale scenarios.

B. Semi-Supervised Learning-Based Schemes

Semi-supervised deep learning has demonstrated high effectiveness in training models with substantial amounts of unlabeled data. These methods are broadly categorized into generative and discriminative approaches. Generative methods aim to model the data generation process, often enhancing generalization ability, while discriminative methods focus on learning the direct relationship between input and output, resulting in higher accuracy. For example, generative methods commonly use techniques such as Gaussian mixture models, naive Bayes classifiers, and deep generative models like variational auto-encoders (VAEs) and generative adversarial networks (GANs) to simulate data generation and create new samples. In contrast, discriminative methods use classifiers like support vector machines, logistic regression, and deep neural networks, including convolutional neural networks, to label unlabeled data directly [20], [21].

To increase the density of fingerprint databases while minimizing construction costs, several semi-supervised assisted methods are utilized to annotate large amounts of unlabeled data with limited labeled data. A semi-supervised classification method called DELM (Semi-supervised Deep Extreme Learning Machine) was proposed in [14]. DELM uses low-cost crowdsourced fingerprint data for training to enhance the classification accuracy for location regions, but does not achieve precise localization. A graph-based semi-supervised manifold alignment method, introduced in [22], preserves neighbor information in both RSS signal space and physical space. Additionally, [23] presents an indoor sub-region positioning scheme that clusters unlabeled RSS fingerprints and associates clusters with indoor sub-regions, generating sub-region fingerprints. However, this approach relies on a detailed indoor topology plan. An unsupervised positioning method, ENAP, was proposed in [7]. It applies unsupervised fusion c-means clustering and calibrated multidimensional scaling to transform unlabeled WiFi trace data into positioning spaces. Additionally, a semi-supervised naive Bayesian method for labeling heterogeneous fingerprints is discussed in [24].

Generative and discriminative approaches are both present in the aforementioned methods and utilize unlabeled data differently. Generative methods are better suited for creating virtual data samples to enhance a model's generalization ability, whereas discriminative methods focus on labeling unlabeled data samples. In WiFi fingerprint labeling, especially with passive crowdsourcing that yields substantial unlabeled

data, a key future challenge is to employ discriminative learning methods to assess the reliability of unlabeled fingerprint data. This will enhance the accuracy of fingerprint locations, improving both the quality and usability of crowdsourced data.

C. Trajectory Matching Schemes

The widespread popularity of mobile devices, equipped with multi-type sensors and robust communication functions, has opened new avenues for constructing fingerprint databases using trajectory matching. Beyond collecting RSS samples, many methods also gather timestamp and sensor data from smartphones to capture information on walking step length, step count, direction, and angular velocity. For instance, a two-dimensional crowdsourced radio map was developed in [8] using an uncertainty-aware WiFi localization module and a tailored Bayesian fusion approach. The WILL system [10] leverages user mobility to eliminate the need for on-site manual fingerprint surveying, enabling automated fingerprint database construction for room-level positioning. However, it suffers from low positioning accuracy. Systems like PiLoc [11] and Pazl [12] use a smartphone's inertial measurement unit (IMU) and the pedestrian dead reckoning (PDR) algorithm to estimate user trajectories. By combining the user's displacement with the RSS signal, these systems achieve low-cost WiFi fingerprint database construction with improved positioning accuracy. However, these systems require prior knowledge of the user's initial position. The LiFS system [25], an evolution of WILL, incorporates smartphone inertial sensors to shift positioning from the physical plane to a high-dimensional fingerprint space, utilizing MDS for physical location labeling. MPiLoc [26], an extension of PiLoc, uses a barometric sensor to cluster trajectories into distinct floors, enabling multi-floor positioning in three-dimensional space. However, relying solely on unlabeled data may introduce errors in model training and compromise positioning accuracy due to RSS signal sensitivity to environmental changes.

In summary, recent advancements in offline fingerprint database construction have provided a solid foundation for location estimation. However, these methods still struggle to achieve satisfactory positioning accuracy due to the limited quantity and reliability of labeled fingerprints in real-world settings. Additionally, they primarily focus on two-dimensional space without incorporating building and floor information. In contrast, our approach employs a semi-supervised multi-task learning framework, providing an efficient method for accurately annotating unlabeled passive crowdsourced fingerprints and constructing a comprehensive multi-scale RSS fingerprint database.

III. PROBLEM STATEMENT

Assuming that $FP_1 = \{(R_n, B_n, F_n, L_n), n = 1, 2, \dots, N_1\}$ is a labeled RSS data set collected in the localization area, $FP_2 = \{(R_n, B_n, F_n, L_n), n = N_1 + 1, \dots, N\}$ is an unlabeled RSS data set collected in the localization area. N_1 and N represent the sizes of the labeled and total datasets respectively, and $N - N_1 \geq N_1$ represents the sizes of the unlabeled data set.

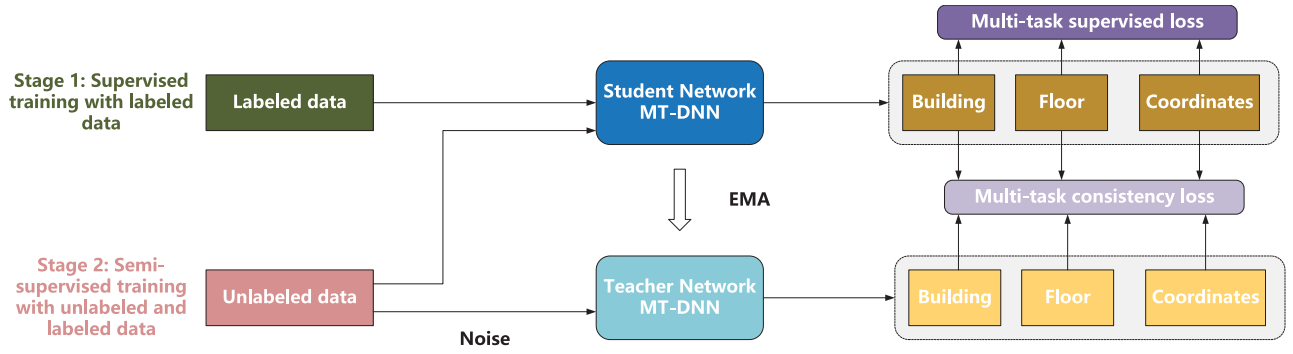


Fig. 3. Model architecture of MTMT-DNN.

Assuming that there are m access points (APs) and a test point i in a given area, the test point can receive RSS from these m APs. Therefore, the fingerprint of point i consists of the RSS values of these m APs, which is described as

$$R_i = (rss_1, rss_2, \dots, rss_m). \quad (1)$$

The corresponding position label of the fingerprint is labeled as

$$Pos_i = (B_i, F_i, L_i), \quad (2)$$

where, $L_i = (x_i, y_i)$ represents the actual physical coordinates, F_i is the floor on which the target is located, and B_i represents the building where the target is located.

Hence, the problem of labeling multi-building multi-floor fingerprints can be defined as accurately annotating passive crowdsourced unlabeled fingerprints using a limited number of labeled fingerprints. Specifically, for a given fingerprint R_i in the unlabeled data set FP_2 , our task is to assign it to the most probable test point i and provide annotations for the building number B_i , floor number F_i , and intra-floor coordinates $L_i = (x_i, y_i)$. Since the building and floor numbers are finite and countable, labeling them can be abstracted as classification tasks. In contrast, the specific horizontal coordinates within a floor are real numbers and not countable, their labeling can be formulated as a regression task.

Based on practical considerations, an inherent relationship exists among the three labeling tasks. Incorrectly labeling the building number will lead to subsequent errors in labeling floor numbers and intra-floor coordinates, resulting in accumulated errors that render the labeling results invalid. Given the interrelated nature of these tasks, their relationships should be fully utilized during model construction. This allows the tasks of labeling building numbers, floor numbers, and intra-floor coordinates to constrain and influence each other, thereby improving the accuracy of cross-scale fingerprint labeling in high-rise consumer-related building environments. This concept aligns with multi-task joint learning in machine learning. By sharing feature parameters and parallel training, related tasks can learn and explore their inherent relationships.

Therefore, the fingerprint labeling problem can be formulated using a multi-task deep learning model, as described in Section IV-B. This model is divided into two classification tasks and a regression task, implemented by jointly learning three interrelated tasks.

IV. MODEL DESIGN

In this section, we provide a detailed description of the designed model.

A. Architecture Overview of MTMT-DNN

We introduce a semi-supervised fingerprint labeling method that leverages a multi-task mean-teacher model. Our objective is to accurately annotate fingerprints by utilizing the information present in the unlabeled data to construct a comprehensive fingerprint database, while also tackling the challenges of missing labels and noise interference. The core concept of our approach involves a multi-task framework incorporating three tasks, building classification, floor classification, and coordinate regression. This framework enhances the robustness and generalization capability of the fingerprint labeling model, and enables the training and adjustment of the model using semi-supervised and unlabeled samples.

The proposed method comprises two stages, illustrated in Fig. 3: 1) supervised training with labeled data, and 2) semi-supervised training that utilizes a combination of labeled and unlabeled data.

During supervised training in stage 1), we initially design a deep neural network labeling model called MT-DNN, based on multi-task learning, to capture the RSS-location mapping relationships for three tasks: building labeling, floor labeling, and intra-floor coordinate labeling. MT-DNN serves as the basic network of both the student network and the teacher network in the MTMT-DNN architecture. In this stage, the model is trained by computing the multi-task supervised loss, which measures the dissimilarity between the predicted results of the unlabeled data and the true labels of the labeled data.

During semi-supervised training in stage 2), we incorporate a mean-teacher mechanism to smooth the model's predictions. We propose a multi-task mean-teacher model, MTMT-DNN, composed of both the student and teacher networks based on MT-DNN. MTMT-DNN includes a supervised multi-task loss function for labeled data, combining the supervised losses of the three tasks. Additionally, it employs a multi-task consistency loss function for unlabeled data, ensuring consistency between the labeling results of the student and teacher networks across all unlabeled data. By jointly training the labeling model using both multi-task supervised loss and consistency loss, this method enhances the performance of

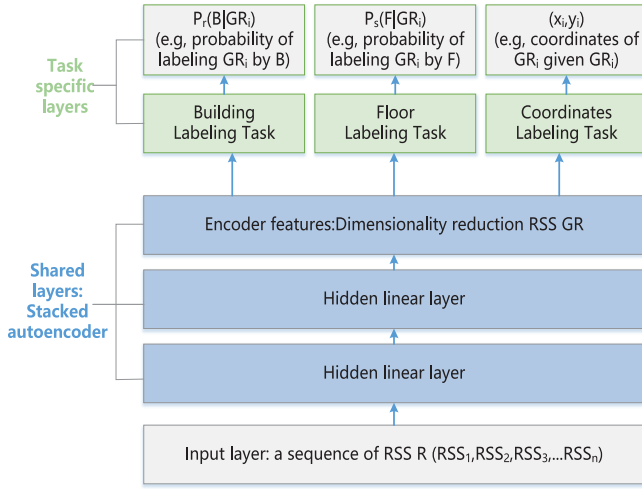


Fig. 4. Network architecture of MT-DNN.

labeling unknown fingerprints, thereby generating a reliable fingerprint database.

B. Multi-Task Deep Learning Fingerprint Labeling Model

Each unlabeled fingerprint in the data set should include information about the building number, floor number, and intra-floor coordinates. The labeling model must estimate all three to determine the precise location of the unlabeled fingerprint. A labeling task is considered complete when the model successfully estimates any of the three pieces of information. Consequently, the fingerprint labeling problem is divided into two classification tasks and a regression task, which are accomplished by simultaneously learning three interrelated tasks. The network model of MT-DNN is depicted in Fig. 4.

The network architecture comprises three layers: an input layer, a shared hidden layer (a stacked autoencoder), and a task layer (three hidden layers and one output layer). The input layer includes 511 nodes, representing the 511-dimensional RSS fingerprint features generated by 511 APs within the building. The shared hidden layer consists of two layers, with 256 and 128 neurons, respectively. The task layer includes a first hidden layer with 256 neurons, a second hidden layer with 128 neurons, a third hidden layer with 64 neurons, and an output layer with 10 nodes. The first classification task uses *ReLU* as the activation function and *Softmax* as the classification loss function, with an output of 3 nodes. Similarly, the second classification task uses *ReLU* as the activation function and *Softmax* as the classification loss function, with an output of 5 nodes. For the regression task, the *tanh* activation function is employed, with *MSE* as the loss function, and the output comprises 2 nodes. The network architecture of the MT-DNN model in Fig. 4 shows that the building, floor, and coordinates labeling tasks share all the neurons before the last hidden layer to explore the intrinsic relationship between the three tasks.

1) *Stacked Autoencoder*: In MT-DNN, a stacked autoencoder is employed before the feedforward classifier to decrease the dimensionality of feature space, mitigate feature noise, and

achieve robust and accurate classification, as depicted in Fig. 4. Two hidden linear layers are used as an encoder to reduce the dimensionality of feature space. The encoded feature sequences for the building and floor classifiers and coordinates regression are obtained using Eq. (3) and Eq. (4), respectively. These sequences serve as the shared feature inputs for the subsequent task layers.

Let $R_i = (rss_1, rss_2, \dots, rss_m)$ represent the RSS fingerprint of a location. The encoded representation of R_i in the first hidden layer is obtained by applying the encoding process,

$$FR(R_i) = f(W_1 R_i + b), \quad (3)$$

and its representation after encoding in the second hidden layer is

$$GR(FR_i) = g(W_2 FR_i + c), \quad (4)$$

where f and g are activation function, *Relu*(), W_1 and W_2 are weight, b and c are bias.

2) *Building and Floor Labeling Tasks*: Given an unlabeled fingerprint GR_i , we employ softmax regression to predict the probability of assigning it a specific building or floor label,

$$P_r(B|GR_i) = \text{softmax}(W_B \cdot GR_i), \quad (5)$$

$$P_s(F|GR_i) = \text{softmax}(W_F \cdot GR_i), \quad (6)$$

and the predicted label for the building or floor is determined based on the label with the highest probability. Here, W_B and W_F are parameter matrices for the specific tasks.

3) *Coordinates Labeling Task*: To accurately locate a specific position within a floor, it is necessary to determine the horizontal and vertical coordinates. This can be achieved by developing a fingerprint-position regression model, represented by Eq. (7),

$$(\hat{x}_i, \hat{y}_i) = W_{r2}[\tanh(W_{r1} GR_i + b_{r1}) + b_{r2}], \quad (7)$$

where W_{r1} and W_{b1} are parameters of the linear function, W_{r2} and W_{b2} are parameters of the regression function.

C. Multi-Task Supervised Loss on Labeled Fingerprints

For the labeled fingerprints $FP_1 = \{(R_n, B_n, F_n, L_n), n = 1, 2, \dots, N_1\}$, each fingerprint consists of input features $R_i = (rss_1, rss_2, \dots, rss_m)$ and a corresponding position label $Pos_i = (B_i, F_i, L_i)$. We consider the annotated building label as the ground truth for the building number (B_i). Similarly, we use the annotated floor label as the ground truth for the floor number (F_i). Additionally, we consider the annotated coordinates label as the ground truth for the coordinates (L_i). Simultaneously, the student network in MT-DNN predicts the label as $Pos_i = (\hat{B}_i, \hat{F}_i, \hat{L}_i)$.

Once the ground truths and predicted labels are obtained, we define the supervised losses for a labeled fingerprint in the building, floor, and coordinates labeling tasks as L_B^δ , L_F^δ , and L_C^δ , respectively,

$$L_B^\delta = - \sum_{i=1}^{N_1} B_i \log \hat{B}_i, \quad (8)$$

$$L_F^\delta = - \sum_{i=1}^{N_1} F_i \log \hat{F}_i, \quad (9)$$

$$L_C^\delta = \sqrt{\frac{1}{N_1} \sum_{i=1}^{N_1} (L_i - \hat{L}_i)^2}. \quad (10)$$

After obtaining the supervised loss of three tasks, the multi-task supervised loss (denote as L^δ) for a labeled fingerprint can be calculated as following two steps.

Initially, we combine the cross-entropy loss functions of building labeling task and floor labeling task, which are of the same magnitude as L_M^δ and defined by Eq. (11). The balance parameter α , with a value between 0 and 1, is used to control the contribution of the two task loss functions.

$$L_M^\delta = \alpha L_B^\delta + (1 - \alpha) L_F^\delta. \quad (11)$$

Moreover, we combine L_M^δ with the loss function L_C^δ of coordinates labeling task to form the fused loss function L^δ , as defined by Eq. (12). The parameters β and λ , each taking a value between 0 to 1, are used to balance the fusion of L_M^δ and L_C^δ .

$$L^\delta = \beta L_M^\delta + \lambda L_C^\delta. \quad (12)$$

D. Multi-Task Consistency Loss on Unlabeled Fingerprints

For unlabeled fingerprints $FP_2 = \{(R_n, B_n, F_n, L_n), n = N_1 + 1, \dots, N\}$, we pass them through the student and teacher networks to obtain the results for three tasks, which are $SPos_i = (SB_i, SF_i, SL_i)$ and $TPos_i = (TB_i, TF_i, TL_i)$. We then enforce consistency between the predictions of the three tasks from the student and teacher networks, resulting in a multi-task consistency loss (L^c),

$$L^c = L_B^c + L_F^c + L_C^c. \quad (13)$$

Here, L_B^c , L_F^c , and L_C^c denote the consistency loss of the building labeling, floor labeling, and coordinates labeling, respectively.

$$L_B^c = \frac{1}{N - N_1} \sum_{i=N_1+1}^N (SB_i - TB_i)^2, \quad (14)$$

$$L_F^c = \frac{1}{N - N_1} \sum_{i=N_1+1}^N (SF_i - TF_i)^2, \quad (15)$$

$$L_C^c = \frac{1}{N - N_1} \sum_{i=N_1+1}^N (SL_i - TL_i)^2. \quad (16)$$

E. Multi-Task Mean-Teacher Semi-Supervised Model Training

We utilize the semi-supervised self-ensembling model in conjunction with multi-task learning to label the positions of multiple buildings and floors. The total loss of our network is

$$L_{total} = \sum_{i=1}^{N_1} L^\delta(R_i) + \varphi \sum_{i=N_1+1}^N L^c(R_j). \quad (17)$$

Here, N_1 and $N - N_1$ represent the number of labeled and unlabeled fingerprints, respectively. $L^\delta(R_i)$ denotes the multi-task supervised loss Eq. (12) for the i -th labeled fingerprint, while $L^c(R_j)$ represents the multi-task consistency loss Eq. (13) for the j -th unlabeled fingerprint. The weight φ balances the multi-task supervised loss on labeled fingerprints and the multi-task consistency loss on unlabeled fingerprints.

We minimize L_{total} to train the student network, and the parameters of the teacher network in each training step are updated using the exponential moving average (EMA) strategy described in [27], [28]. The parameters of the teacher network at the t -th training iteration are

$$\theta'_t = \eta \theta'_{t-1} + (1 - \eta) \theta_t, \quad (18)$$

where θ_t denotes the student network parameters at the t -th training iteration. The EMA decay η is empirically set as 0.95 according to experiments.

F. Evaluation of Predicted Unlabeled Fingerprints Quality

We assess the reliability of unlabeled fingerprints using a mechanism for evaluating the quality of our model. This evaluation consists of two steps, i.e., multi-task supervised model quality evaluation and multi-task semi-supervised model quality evaluation.

The quality of the multi-task supervised model (student network) is assessed using the labeled test data set. During the training of the multi-task supervised model, the labeled data is input into the student network to obtain a trained model. Consequently, in training with labeled data, we aim to develop a high-quality student network model to guide the training of the teacher network and enhance the prediction accuracy for the unlabeled data. The high quality of the student network model is evidenced by the close correspondence between the model's predictions on the test data set and the true labels. This alignment is achieved by minimizing the multi-task supervised loss function Eq. (12).

The quality of the multi-task semi-supervised model is evaluated based on the unlabeled data set. During training, we obtain a teacher network model by applying the exponential moving average strategy to the trained student network model. Subsequently, during the training with unlabeled data, both the student and teacher network models predict the unlabeled data, and their prediction labels are compared to evaluate the models' performance on the unlabeled data set. When the unlabeled data is of high quality, the prediction labels of the student model closely align with those of the teacher network on the unlabeled data set. This alignment is achieved by minimizing the multi-task consistency loss function Eq. (13).

Hence, it can be concluded that the mean-teacher semi-supervised labeling model achieves superior performance and provides more reliable label predictions when the total loss Eq. (17) is minimized.

V. EXPERIMENTAL ANALYSIS

A. Experiment Setups

1) *Experiment Sites*: The campus teaching and office building scenario represents a typical complex environment with

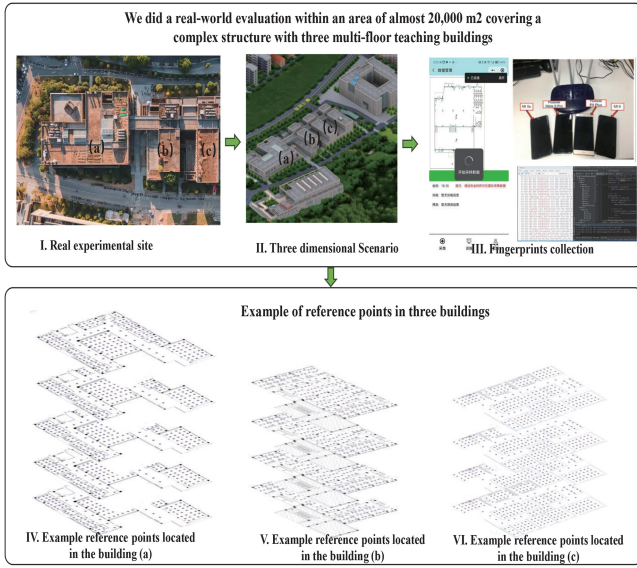


Fig. 5. Real experimental scenarios and example of reference points.

multiple buildings and floors. To objectively evaluate the positioning performance of MTMT-DNN in real-world environments, we tested it in a large-scale teaching and office complex covering three adjacent buildings. Building (a) is a five-floor teaching building, Building (b) is a four-floor teaching building, and Building (c) is a four-floor office building. The varying levels of congestion in these buildings contribute to the complexity of the environment. The three buildings are physically adjacent and connected by corridors. Each building has a unique overall structure, creating different scenarios: Building (a) has a large open space in the center, while Buildings (b) and (c) feature relatively long and narrow corridors. Each building has floors with partially identical structures. The total coverage area of the experimental scenario is 20,000 square meters. The experimental setup, including the real-world scene, three-dimensional structure, collection tools, floor plan structure, and distribution of reference points, is shown in Fig. 5. Considering the characteristics and dimensions of the spaces, we carefully chose between 3 to 20 reference points (RPs) within various enclosed areas, ensuring that the distance between each pair of RPs was approximately 3-5 meters.

Additionally, a total of 13 surveyors equipped with five different smartphone models (Huawei P9 Plus, MI 5s, Huawei Mate 9 Pro, MI 6, Huawei P40 Pro) collected training fingerprint samples for each RP. To collect the fingerprint sets and evaluate the positioning performance, we utilized a WeChat Mini Program developed by our laboratory. If APs' signals were not received, we set the RSS readings to -100 dBm.

2) *Fingerprint Database and Model Parameters*: The fingerprint database collected in the large-scale teaching and office buildings consists of 11,250 RSS fingerprints obtained from the enclosed spaces of buildings (a), (b), and (c). The three buildings contributed 4,250, 3,600, and 3,400 fingerprints, respectively. During the testing phase, the collected fingerprints are divided into an unlabeled set and a labeled

TABLE I
PARAMETERS OF THE STUDENT NETWORK OF MTMT-DNN

	Parameters	Value
Learning Parameters	Iterations	3000
	Batch Size	512
	Optimizer	Adam
	Dropout	0.005
	φ	0.7
	η	0.95
Shared layers	Input	511
	Hidden layer	511-256-128
	Output	128
Task specific layers	Hidden layer	256-128-64
	Building/Floor labeling activation	Relu
	Building/Floor labeling loss	Softmax
	Coordinates labeling activation	Tanh
	Coordinates labeling loss	MSE
	Output	10

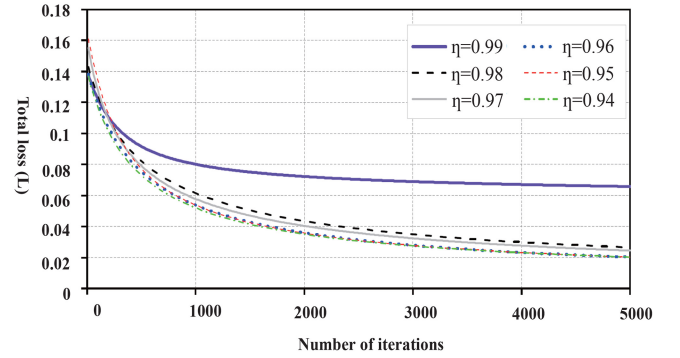


Fig. 6. Total loss on different iterations.

set in an 8:2 ratio. Each fingerprint comprises 515 attributes, including 511 RSS features and 4 location labels. The location labels encompass a building number B , a floor number F , and a pair of x and y -coordinates.

Moreover, the student network structures and learning-related values in the experiments are listed in Table I. The effectiveness of the MTMT-DNN is verified based on the above experiment setting.

B. Reliability Evaluation of Predicted Unlabel Fingerprints

The optimization objective of the MTMT-DNN model is to minimize the overall loss to obtain highly reliable predicted labels. This is achieved by training a dependable student prediction model using the multi-task supervised loss function. Additionally, the multi-task consistency loss function is employed between the student and teacher models to filter out unreliable predictions in the semi-supervised loss. The teacher model guides the training of the student model, resulting in more reliable prediction results. Fig. 6 illustrates the loss function of the MTMT-DNN model after 5,000 iterations for various η values. From Fig. 6, it is evident that the total loss decreases and the predicted labels become progressively more reliable during the training process. Consequently, the usability

TABLE II
COMPARISON OF BUILDING AND FLOOR LABELING ACCURACY BETWEEN FINGERPRINTS ANNOTATION MODELS

	Learning mode	Task learning mode	Feature sharing	Building/floor labeling model	Coordinates labeling model	Building labeling accuracy(%)	Floor labeling accuracy (%)
MTMT-DNN	Semi-supervised	Multit-task	Yes	Softmax	MLP	99-100	99
DNN	Supervised	Single-task	No	MLP	MLP	<80	<76
CNN	Supervised	Single-task	No	1D-CNN	MLP	<80	<78
MT-DNN	Supervised	Multit-task	Yes	Softmax	MLP	<82	<80
SCNN	Semi-supervised	Single-task	No	1D-CNN	1D-CNN	<90	<85
SSAE	Semi-supervised	Single-task	No	encoder-decoder	encoder-decoder	<93	<88
SSVM	Semi-supervised	Single-task	No	SVM	SVR	<76	<70

of the annotated fingerprint database is enhanced. The optimal performance is attained when η takes a value of 0.95.

C. Building and Floor Labeling Accuracy Evaluation

We consider several different network compositions that combine classification and regression sub-tasks to achieve optimal performance. Specifically, we compare the MTMT-DNN model with six other fingerprint annotation models that utilize distinct learning methods and task training approaches. These models include the Deep Neural Network (DNN), Convolutional Neural Network (CNN), Multi-Task Deep Neural Network (MT-DNN), Semi-Supervised CNN (SCNN), Semi-Supervised Autoencoder (SSAE), and Semi-Supervised SVM (SSVM). Among these models, DNN and CNN are supervised, single-task models, MT-DNN is a supervised, multi-task model, while SCNN, SSAE, and SSVM are semi-supervised, single-task models. The specific learning methods, task training approaches, and building/floor labeling performance of each model are presented in Table II.

Initially, the experiment compares and analyzes the MTMT-DNN model alongside the supervised, single-task models DNN and CNN. In the supervised, single-task models, the building and floor annotation models are trained independently using labeled data. Subsequently, unlabeled data is used as test data and fed into the trained building and floor annotation models for labeling. However, this approach neglects cumulative errors and the valuable information present in the unlabeled data set, leading to diminished accuracy in label predictions.

Subsequently, a comparison and analysis were performed between the MTMT-DNN model and the supervised, multi-task model MT-DNN. In MT-DNN, the building and floor label annotation models are jointly trained using labeled data, and the interdependency between the two tasks is explored through shared hidden layers. Unlabeled data is employed as test data for prediction. However, due to the considerably smaller quantity of labeled data compared to unlabeled data, the model's labeling accuracy is not high, achieving only 82% and 80%.

Furthermore, a comparison and analysis were conducted between the MTMT-DNN model and the semi-supervised, single-task models SCNN, SSAE, and SSVM. First, independent models for building and floor label annotation were trained using labeled data, and subsequently, these models

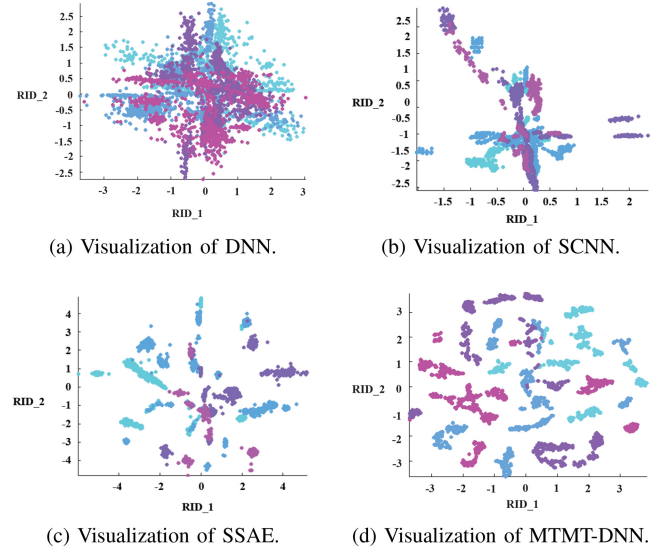


Fig. 7. Visualization of floor labeling results in building (b).

were utilized to predict the unlabeled data. The experimental results revealed that the semi-supervised, single-task methods CNN, autoencoder, and SVM did not perform well in building and floor recognition. SSVM achieved a maximum accuracy of 76% for building annotation, SCNN achieved 90%, and SSAE achieved 93%. For floor annotation, SSVM achieved a maximum accuracy of 70%, while SCNN achieved 85% and SSAE achieved 88%.

D. Floor Labeling Results Visually Evaluation

To visually compare the effectiveness of label annotation, the experiment employed dimensional reduction visualization tools, specifically t-SNE, and used the accuracy of floor annotation as the evaluation metric. The labeled fingerprints were visually analyzed to evaluate the accuracy of the annotations.

We initially analyzed and compared the labeling performance of DNN, SCNN, SSAE, and MTMT-DNN using fingerprints collected in building (b). We applied t-SNE to reduce the original RSS feature space and its projected counterpart to two dimensions. The results are plotted in Fig. 7(a), Fig. 7(b), Fig. 7(c), and Fig. 7(d) with different colors representing the floors. In Fig. 7(a), the fingerprints labeled by DNN exhibit overlapping distributions, even for samples collected from different floors, resulting in suboptimal

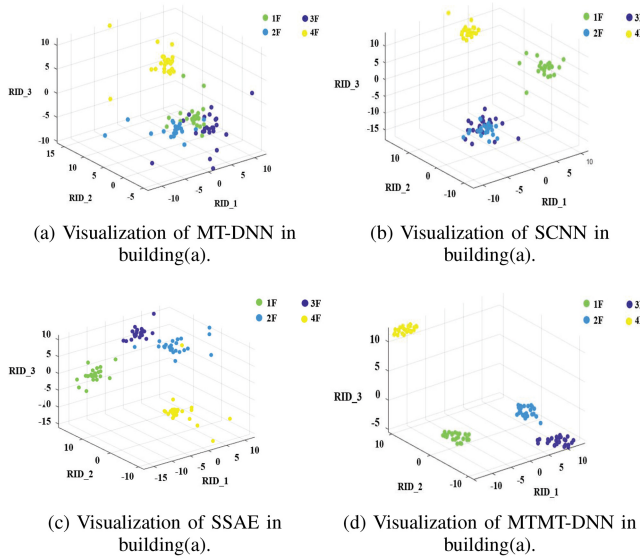


Fig. 8. Visualization of floor labeling results in building (a).

labeling results. In Fig. 7(b) and Fig. 7(c), the distinguish ability of fingerprint location labels labeled by SCNN and SSAE is significantly higher than those of DNN, but it still cannot meet the accuracy requirements of practical scenarios. Conversely, in Fig. 7(d), after being labeled by the proposed MTMT-DNN, the fingerprints from different floors exhibit clear classification boundaries. Therefore, we can conclude that the MTMT-DNN model effectively utilizes the location information present in both the labeled and unlabeled fingerprints, resulting in excellent annotation performance.

We also conducted tests and comparisons to assess the labeling performance of MT-DNN, SCNN, SSAE, and MTMT-DNN using fingerprints collected from three multi-floor buildings. The architecture of each building exhibits variations, thus constituting distinct scenarios. We randomly selected 10% of the samples from different buildings as a test set for evaluation. The first three most representative features were utilized to visualize the classification results, as shown in Fig. 8, Fig. 9, and Fig. 10. These results demonstrate the substantial difference in floor labeling capabilities between MT-DNN, SCNN, SSAE, and MTMT-DNN. In each building, the results of MT-DNN show overlaps. Although it can also achieve floor labeling, there are no clear boundaries between different categories due to the small number of training samples. While SCNN and SSAE achieves floor labeling to some extent, the results also exhibit overlapping areas with no clear boundaries. In contrast, the boundaries between different floors labeled by MTMT-DNN are clear. Therefore, compared to supervised multi-tasking mode and semi-supervised single-task mode, the semi-supervised multi-tasking mode makes full use of the unlabeled data, resulting in more reliable labeling outcomes.

E. Coordinates Labeling Accuracy Evaluation

To test and compare the accuracy of coordinates labeling using different learning modes, we randomly selected 20% of the RPs from each building. Among these RPs, 80% consisted

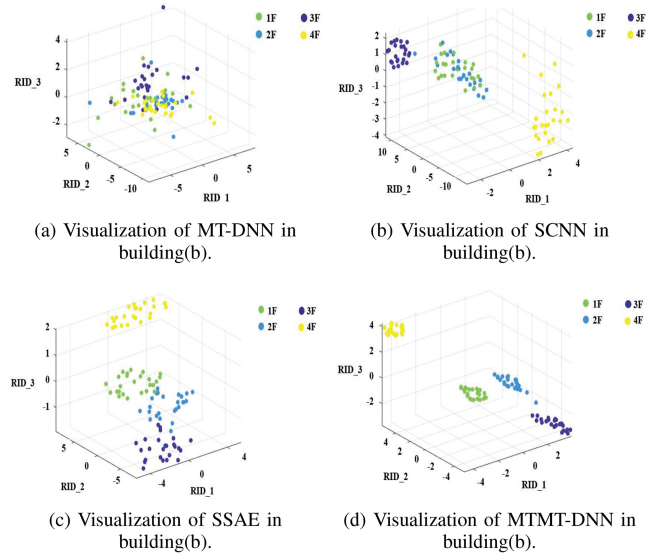


Fig. 9. Visualization of building and floor labeling results in building (b).

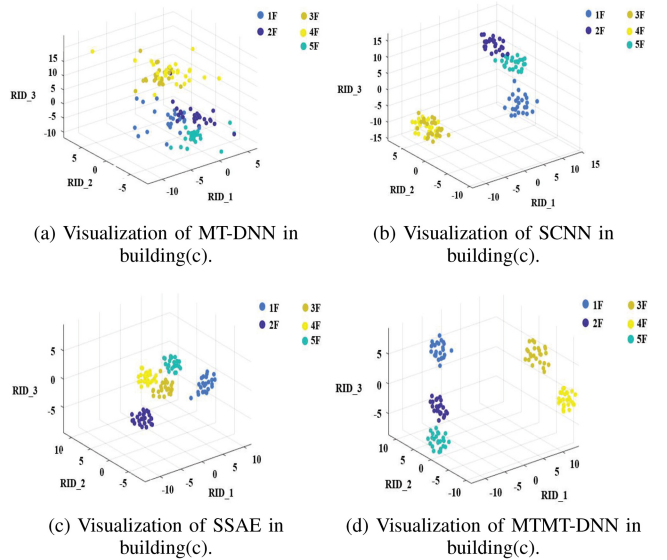


Fig. 10. Visualization of building and floor labeling results in building (c).

of unlabeled fingerprint data, while the remaining 20% were labeled fingerprints.

To examine the effect of semi-supervised learning approaches on coordinate labeling performance, we compared the semi-supervised, multi-task model MTMT-DNN with three semi-supervised single-task models: SSAE, SCNN, and SSVM. For each test point, these models were jointly trained on both the labeled and unlabeled fingerprints in the test data set to estimate the location coordinates of the unlabeled fingerprints. The experiment was repeated 30 times under each condition to verify the accuracy of coordinate labeling. The average labeling errors for the two conditions over 30 iterations are depicted in Fig. 11. The orange bars represent MTMT-DNN, the blue bars represent SSAE, the green bars represent SCNN, and the gray bars represent SSVM. The results indicate that the average labeling error of MTMT-DNN ranges from 2 m to 4 m, SSAE ranges from 4 m to 8 m,

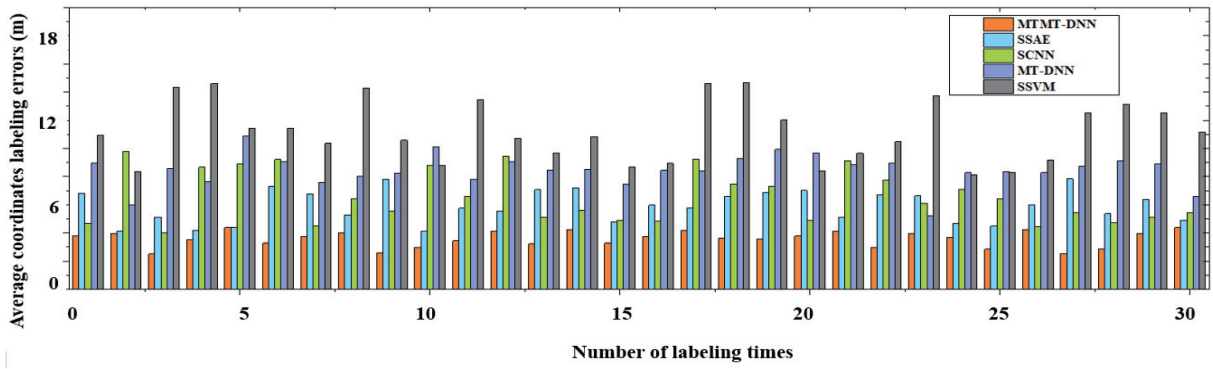


Fig. 11. Comparison of coordinates labeling accuracy of annotation models using different learning mode.

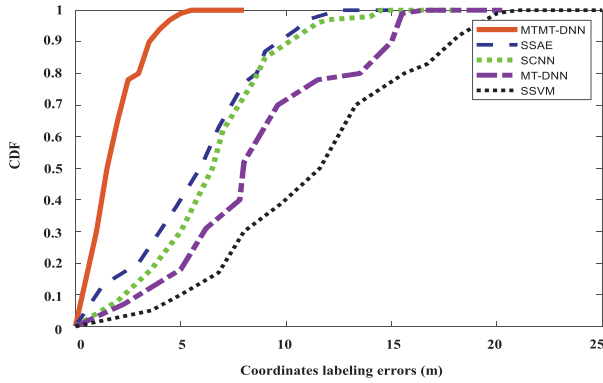


Fig. 12. Comparison of CDF of annotation models using different learning mode.

SCNN ranges from 4 m to 10 m, and SSVM ranges from 8 m to 15 m. The experimental results suggest that the annotation error using MTMT-DNN is significantly lower compared to other semi-supervised single-task models, and the annotation performance of MTMT-DNN demonstrates greater stability with less error fluctuation.

Finally, to clearly visualize the cumulative probability of coordinate annotation errors across different numerical ranges and further evaluate the stability and robustness of different annotation models, we statistically analyzed and compared the labeling error distributions of MT-DNN, SSVM, SSAE, SCNN, and MTMT-DNN. As shown in Fig. 12, 90% of the cumulative positioning errors of MTMT-DNN are within 5 m, which is approximately 10 m lower than MT-DNN, 13 m lower than SSVM, 4 m lower than SSAE, and 5 m lower than SCNN. This clearly demonstrates that the proposed multi-task semi-supervised model has significant advantages compared to the multi-task supervised model and the single-task semi-supervised models. When the amount of labeled data is far less than the amount of unlabeled data, the annotation model proposed in this paper exhibits better processing capabilities and stability.

VI. DISCUSSION

A. Comparison With Existing Approaches

As reviewed in Section II, some studies have employed interpolation, semi-supervised learning, or trajectory matching

to annotate fingerprint location label and construct a comprehensive fingerprint database. However, due to the substantial imbalance between the number of labeled and unlabeled fingerprints collected in real-world scenarios and the sparse distribution of fingerprint RPs, these databases are often limited in both quantity and reliability. Furthermore, these techniques primarily focus on 2D spaces and do not incorporate building or floor information relevant to real-world applications. In contrast, the proposed MTMT-DNN significantly outperforms existing positioning techniques in the following three aspects: reducing manual site survey efforts, enhancing fingerprint location labeling reliability, and improving model feasibility for fingerprint location labeling.

- *Reducing manual effort of site surveys.* Unlike traditional methods requiring large amounts of labeled data, MTMT-DNN aims to maximize the utility of unlabeled fingerprint samples, achieving high-quality fingerprint labeling with minimal labeled data. This approach not only reduces the manual effort needed for data collection and labeling but also enhances the density and comprehensiveness of the fingerprint database.
- *Enhancing fingerprint location labeling reliability.* Differing from current labeling accuracy influenced by building structure and data imbalance, MTMT-DNN first addresses these challenges by developing a multi-task labeling model that fully explores the relationships among building, floor, and intra-floor coordinates. This model enables mutual constraints across building, floor, and intra-floor coordinate labeling, reducing cumulative errors in multi-scale location labels in complex building environments. Additionally, a multi-task supervised loss function and a multi-task consistency loss function are designed to assess the reliability of predicted labels, ensuring more accurate and reliable fingerprint labeling.
- *Improving model feasibility for fingerprint location labeling.* Unlike existing labeling schemes that focus only on 2D space, MTMT-DNN employs a multi-task joint learning and semi-supervised mean-teacher framework, making it well suited for real-world multi-building and multi-floor environments. Moreover, this approach efficiently utilizes large amounts of unlabeled data with minimal manual effort. Both make our solution more feasible and easier to implement in real-world engineering

applications and applicable to other consumer electronics environments.

B. Potential Limitations and Improvements in More Complex Environments

The proposed MTMT-DNN mainly consider indoor environments in daily consumer electronics scenarios, such as office buildings and shopping malls. However, in more complex scenarios, such as irregularly distributed buildings (e.g., hospitals and factories), or disaster situations (e.g., fires situations with sparse distributed APs and high time variability), the approach may not achieve high location labeling accuracy. This limitation arises from the lack of mechanisms for updating and adapting the fingerprint database and multi-task labeling model to a new and complex environment. Consequently, we summarize the potential limitations and improvements of our proposal in more complex consumer electronics environments as follows:

- *How to design a fingerprint database update mechanism to adapt to new environments?* In complex consumer electronics scenarios, WiFi signals vary dynamically and are affected by factors such as weather, temperature, building structure, internal layout, and positioning equipment. These environmental influences can shift fingerprint features across scenarios, impacting positioning accuracy. In the future, our focus will be on we aim to ensuring the quantity and quality of Cross environmental fingerprint database by integrating unsupervised and transfer learning techniques to enable feature adaptation and database updates in new sites.
- *How to design a cross-environment multi-task adaptive partitioning and learning mechanism to ensure the scalability of labeling model?* The core idea of the proposed MTMT-DNN is to enhance single-task performance by simultaneously learning multiple related tasks within the same site. In contrast, meta-learning focuses on training models to rapidly adapt to new tasks. In new environments, it is essential to consider the distinct structural characteristics of buildings within complex consumer electronics settings and adjust the labeling model accordingly. To address this need, our future efforts will focus on leveraging meta-learning to enable models to automatically learn hyper parameters and adapt to new environments by utilizing knowledge from similar labeling tasks in prior settings.

VII. CONCLUSION

In this paper, we are dedicated to resolving the absence of WiFi RSS fingerprint data in consumer electronics-related commercial positioning services. By leveraging a small amount of labeled fingerprint data to label large amounts of unlabeled crowdsourced fingerprints, we can efficiently and conveniently construct an offline fingerprint database. This approach addresses the challenges of migrating WiFi fingerprint databases from laboratory environments to complex consumer electronics environments.

Specifically, we present a novel semi-supervised multi-task WiFi RSS fingerprints annotation model using a mean-teacher network framework. The annotation model generates accurate annotations based on a limited number of labeled WiFi RSS fingerprints and a large set of unlabeled fingerprints. Our approach involves designing a multi-task network that can simultaneously predict building, floor, and coordinate labels for unlabeled fingerprints collected through passive crowdsourcing, leveraging their intrinsic information. Subsequently, we apply semi-supervised learning to utilize a large number of additional unlabeled fingerprints, further enhancing the labeling performance. We evaluate our model using various network compositions that integrate classification and regression sub-tasks, employing a data set collected from three neighboring multi-story buildings. The results demonstrate its competitive performance, indicating that our model can maintain accurate location labels annotation even with a limited number of labeled fingerprints. This effectively reduces the need for labor-intensive site surveys and enhances the reliability of fingerprint data.

In the future, we will explore using unsupervised learning and meta learning techniques to enhance the robustness and universality of fingerprint databases and labeling models for various consumer electronics environments.

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